

1. Administrative & Study Identification

Full Study Title: An Open-label, Phase 1, Multicenter Study to Evaluate the Safety and Preliminary Anti-tumor Activity of NT-112 in Human Leukocyte Antigen-C08:02-Positive Adult Subjects With Unresectable, Advanced, and/or Metastatic Solid Tumors That Are Positive for the KRAS G12D Mutation
Short Title/Acronym: Phase 1 KRAS TCRT in Advanced Solid Tumors
Protocol Number: NT-112-301
Secondary IDs: NCI-2024-01226, NEOGENE-NT-112-301
NCT Number: NCT06218914
Sponsor Name: AstraZeneca
Investigational Product: NT-112 (Autologous, engineered T Cells targeting KRAS G12D / KRAS TCRTs)
Phase of Development: Phase I
Medical Monitor: AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective
* **Primary Objective:** To evaluate the safety, preliminary anti-tumor activity, and to determine the incidence of Dose-Limiting Toxicities (DLTs), Treatment-Emergent Adverse Events (TEAEs), and Serious Adverse Events (SAEs).
* **Secondary Objectives:** To evaluate the Objective Response Rate (ORR) per RECIST v1.1, Duration of Response (DOR), Progression-Free Survival (PFS), Overall Survival (OS), Clinical Benefit Rate (CBR), Time to Response (TTR), and Best Overall Response (BOR).

2.2 Background & Rationale
To address the significant unmet medical need in patients with unresectable, advanced, and/or metastatic solid tumors (such as NSCLC, pancreatic, colorectal, and endometrial adenocarcinomas) whose tumors harbor a KRAS G12D variant mutation. Standard cytotoxic and targeted therapies have limited efficacy in these advanced settings after ≥ 1 line of approved systemic standard of care (SOC) has failed. This master protocol investigates TCR-Engineered T cells (TCRTs) capable of recognizing and targeting KRAS mutations in subjects with specific HLA types (HLA-C*08:02, HLA-A*11:01, or HLA-A*11:02) to provide a novel, potentially curative adoptive cell therapy option.

3. Study Design & Methodology

3.1 Design Framework
* **Study Type:** Interventional
* **Control Type:** Single-arm
* **Blinding:** Open-label
* **Randomization:** N/A (Dose-escalation/expansion master protocol)

3.2 Planned Enrollment & Duration
* **Target N:** Approximately 24 to 108 evaluable patients
* **Number of Sites:** Multi-center

(Multiple states across the United States including CA, FL, IL, KS, MA, MO, NY, PA, TX) * **Estimated Study Start:** 22-Mar-2024 * **Estimated Primary Completion:** 30-Aug-2027 (Note: Estimated long-term Study Completion Date is 18-Nov-2043 for standard gene therapy follow-up)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years (All Genders). 2. Diagnosed with NSCLC, Colorectal adenocarcinoma, Pancreatic adenocarcinoma, Endometrial Cancer, or any other solid tumor. 3. Tumors must harbor a KRAS G12D variant mutation and subject must be HLA-C*08:02 positive, HLA-A*11:01, or HLA-A*11:02 positive in at least one allele. 4. Subject has advanced solid cancer, defined as unresectable, advanced, and/or metastatic disease (Stage III or IV) after at least 1 line of approved systemic standard of care (SOC) treatment regimen, with no available curative treatment options. 5. Presence of at least 1 measurable lesion per RECIST v1.1. 6. Eastern Cooperative Oncology Group (ECOG) performance status of 0 to 1 at the time of enrollment.

4.2 Exclusion Criteria 1. Any other primary malignancy within the 3 years prior to enrollment (except for non-melanoma skin cancer, carcinoma in situ or low-grade prostate cancer). 2. Known, active primary central nervous system (CNS) malignancy. 3. History of prior adoptive cell and gene therapy, allogeneic stem cell transplant, or solid organ transplantation. 4. History of stroke or transient ischemic attack within the 12 months prior to enrollment. 5. History of clinically significant cardiac disease within 6 months prior to enrollment or heart failure at any time prior to enrollment. 6. Systemic therapy within at least 2 weeks or 3 half-lives, whichever is shorter, prior to enrollment. 7. Any form of primary immunodeficiency. 8. Active immune-mediated disease requiring systemic steroids or other immunosuppressive treatment (except if related to prior checkpoint inhibitor therapy). 9. Female of childbearing potential who is lactating or breast feeding at the time of enrollment. 10. Prior treatment with pan-KRAS or KRAS G12D targeting agents (unless presence of KRAS G12D mutation is re-confirmed after completion of said treatment).

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Single infusion of TCR-Engineered T cells (KRAS TCRTs / NT-112) following a dose-escalation schedule. * **Route of Administration:** Intravenous (IV) Infusion. * **Duration of Treatment:** Single administration, with subsequent acute monitoring and long-term observational follow-up (up to 15 years).

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard supportive care for advanced oncology patients and management of immune-related toxicities (e.g., tocilizumab/steroids for Cytokine Release Syndrome). * **Prohibited:** Concurrent investigational agents, systemic immunosuppressants (except for managing AE/DLTs), or additional anti-cancer systemic therapy without recording a disease progression event.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, HLA/KRAS molecular testing, Baseline Imaging, RECIST assessment
Baseline	Day 0	Lymphodepletion (if protocolized), TCRT Infusion (First Dose), Vital Signs
Interim	Weeks 1 to 4	Intensive Safety Labs (DLT window), CRS/ICANS monitoring, Efficacy Assessment
End of Study	Up to 15 Years	Final Vital Signs, Exit Interview, Long-term Gene Therapy Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Incidence of Dose-Limiting Toxicities (DLTs), Treatment-Emergent Adverse Events (TEAEs), and Serious Adverse Events (SAEs). * **Measurement Method:** Evaluated using NCI Common Terminology Criteria for Adverse Events (CTCAE) and clinical safety assessments.

7.2 Secondary & Exploratory Endpoints * Objective Response Rate (ORR) per RECIST v1.1 * Progression-Free Survival (PFS) and Overall Survival (OS) * Duration of Response (DOR), Time to Response (TTR), and Clinical Benefit Rate (CBR)

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous monitoring of patient clinical status, particularly looking for Cytokine Release Syndrome (CRS) and Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS).
 - **Serious Adverse Events (SAEs):** Routine 24-hour reporting to the Sponsor and regulatory bodies.
 - **Data Monitoring Committee (DMC):** An independent internal/external safety committee will review dose-escalation cohorts and clear subsequent dose levels.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Safety Set (all patients who receive the TCRT infusion) and Efficacy Evaluable Set (patients with measurable disease meeting follow-up criteria).
 - **Sample Size Justification:** Expected $N=24$ to 108 ; guided by dose-escalation rules (e.g., standard 3+3 or similar advanced design) designed to find the Maximum Tolerated Dose (MTD), followed by dose expansion cohorts.
 - **Handling of Missing Data:** Efficacy missing data typically censored at the last evaluable timepoint; non-evaluable patients during the DLT window will be replaced.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** High-frequency onsite monitoring during the initial DLT period, scaling to remote and routine follow-ups.
 - **Audit Trail:** Strict chain of identity and chain of custody tracking for the autologous cellular product, alongside standard eCRF data clarification structures.
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11. Study Identifiers & Governance

- **Primary ID:** NT-112-301
 - **Secondary IDs:** NCI-2024-01226, NEOGENE-NT-112-301
 - **Registry Number:** NCT06218914
 - **Sponsor/Lead Organization:** AstraZeneca
 - **Study Title:** Phase 1 Study to Investigate TCRTs KRAS Mutation in Unresectable, Advanced, and/or Metastatic Solid Tumors
 - **Official Regulatory Title:** An Open-label, Phase 1, Multicenter Study to Evaluate the Safety and Preliminary Anti-tumor Activity of NT-112 in Human Leukocyte Antigen-C*08:02-Positive Adult Subjects With Unresectable, Advanced, and/or Metastatic Solid Tumors That Are Positive for the KRAS G12D Mutation
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12. Clinical Framework (Solid Tumor Specific)

- **Primary Condition:** Unresectable, Advanced, and/or Metastatic Solid Tumors (NSCLC, Colorectal, Pancreatic, Endometrial)
- **Diagnosis Threshold:** Confirmed KRAS G12D variant mutation and positive for HLA-C*08:02, HLA-A*11:01, or HLA-A*11:02
- **Medical Methodology:** Adoptive Cell Therapy / Immunotherapy (TCR-Engineered T cells)
- **Optimization Target:** Eradication/Reduction of tumor burden via T-cell mediated destruction of KRAS G12D expressing cells

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Role	Name	Signature	Date		:--	:--	:--	:--
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]					
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]					